

# iClusterBayes Summary Report

Automated Report

## Hyperparameters

```
cat('sdev = 0.005, beta.var.scale = 0.8\n')
```

```
## sdev = 0.005, beta.var.scale = 0.8
```

## Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.5339  0.7894  0.8311  0.8104  0.8456  0.8817

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.09167 0.13833 0.13373 0.17111 0.90111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.09444 0.14167 0.13696 0.17556 0.69333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.007778 0.025556 0.158427 0.184444 1.000000
```

```

Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0055556 0.0094444 0.0884393 0.0351389 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.007778 0.031111 0.199776 0.260556 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.5261  0.6944  0.7339  0.7229  0.7617  0.8111

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04611 0.09278 0.09781 0.13389 0.83611
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04778 0.09667 0.10043 0.13889 0.63722
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0283333 0.3466888 0.8361111 1.0000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0027778 0.2025574 0.1670833 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.07167 0.82167 0.61118 0.96111 1.00000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.8839  0.9911  0.9939  0.9929  0.9961  0.9994

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03056  0.06944  0.08765  0.11556  0.87611
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03222  0.07222  0.08907  0.12000  0.65222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000  0.0000000  0.0001767  0.0000000  0.5055556
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000  0.0000000  0.0001906  0.0000000  0.0594444
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000  0.0000000  0.0001198  0.0000000  0.0316667
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000  0.0000000  0.0001001  0.0000000  0.1977778

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9600  0.9922  0.9944  0.9938  0.9961  1.0000

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01833  0.04583  0.06228  0.08278  0.82722
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

| Modality 1 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.02000 | 0.04778 | 0.06414 | 0.08667 | 0.75722 |

| Modality 2 : |           |           |           |           |           |
|--------------|-----------|-----------|-----------|-----------|-----------|
| Min.         | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      |
| 0.000e+00    | 0.000e+00 | 0.000e+00 | 3.162e-05 | 0.000e+00 | 2.156e-01 |

| Modality 3 : |           |           |           |           |           |
|--------------|-----------|-----------|-----------|-----------|-----------|
| Min.         | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      |
| 0.000e+00    | 0.000e+00 | 0.000e+00 | 1.353e-05 | 0.000e+00 | 1.722e-02 |

| Modality 4 : |           |           |           |           |           |
|--------------|-----------|-----------|-----------|-----------|-----------|
| Min.         | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      |
| 0.000e+00    | 0.000e+00 | 0.000e+00 | 1.382e-05 | 0.000e+00 | 3.889e-03 |

| Modality 5 : |           |           |           |           |           |
|--------------|-----------|-----------|-----------|-----------|-----------|
| Min.         | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      |
| 0.0000000    | 0.0000000 | 0.0000000 | 0.0001802 | 0.0000000 | 1.0000000 |

--- Summary for K = 5 ---

Z.ar:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.4994 | 0.6383  | 0.6900 | 0.6869 | 0.7400  | 0.8528 |

beta.ar:

| Modality 1 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.01222 | 0.03333 | 0.04799 | 0.06556 | 0.81278 |

| Modality 2 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 3 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 4 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 5 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

| Modality 1 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.01333 | 0.03556 | 0.04962 | 0.06889 | 0.53778 |

| Modality 2 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.00000 | 0.00000 | 0.01101 | 0.00000 | 1.00000 |

| Modality 3 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 0.8189  | 0.9144 | 0.8828 | 0.9983  | 1.0000 |

| Modality 4 : |  |  |  |  |  |
|--------------|--|--|--|--|--|
|--------------|--|--|--|--|--|

| Min.     | 1st Qu.  | Median   | Mean     | 3rd Qu.  | Max.     |
|----------|----------|----------|----------|----------|----------|
| 0.000000 | 0.000000 | 0.000000 | 0.005507 | 0.000000 | 0.381111 |

Modality 5 :

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.0000 | 0.0000  | 0.0000 | 0.0247 | 0.0000  | 0.8050 |

--- Summary for K = 6 ---

Z.ar:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.5128 | 0.6889  | 0.7367 | 0.7326 | 0.7817  | 0.8711 |

beta.ar:

Modality 1 :

| Min.     | 1st Qu.  | Median   | Mean     | 3rd Qu.  | Max.     |
|----------|----------|----------|----------|----------|----------|
| 0.000000 | 0.009444 | 0.025000 | 0.039897 | 0.052222 | 0.857778 |

Modality 2 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 3 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 4 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 5 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

Modality 1 :

| Min.     | 1st Qu.  | Median  | Mean    | 3rd Qu. | Max.    |
|----------|----------|---------|---------|---------|---------|
| 0.000000 | 0.010000 | 0.02667 | 0.04089 | 0.05486 | 0.43222 |

Modality 2 :

| Min.      | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      |
|-----------|-----------|-----------|-----------|-----------|-----------|
| 0.0000000 | 0.0000000 | 0.0000000 | 0.0004177 | 0.0000000 | 0.6261111 |

Modality 3 :

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.0000 | 0.4128  | 0.6367 | 0.5873 | 0.7989  | 1.0000 |

Modality 4 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 0    | 0       | 0      | 0    | 0       | 0    |

Modality 5 :

| Min.      | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      |
|-----------|-----------|-----------|-----------|-----------|-----------|
| 0.0000000 | 0.0000000 | 0.0000000 | 0.0007404 | 0.0000000 | 0.6816667 |

--- Summary for K = 7 ---

Z.ar:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.9189 | 0.9900  | 0.9928 | 0.9921 | 0.9944  | 0.9994 |

beta.ar:

Modality 1 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
|------|---------|--------|------|---------|------|

0.000000 0.008333 0.021111 0.041498 0.047778 0.866667

Modality 2 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 3 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 4 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 5 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

Modality 1 :

| Min.     | 1st Qu.  | Median   | Mean     | 3rd Qu.  | Max.     |
|----------|----------|----------|----------|----------|----------|
| 0.000000 | 0.008889 | 0.022222 | 0.040849 | 0.051111 | 0.586667 |

Modality 2 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 0    | 0       | 0      | 0    | 0       | 0    |

Modality 3 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 0    | 0       | 0      | 0    | 0       | 0    |

Modality 4 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 0    | 0       | 0      | 0    | 0       | 0    |

Modality 5 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 0    | 0       | 0      | 0    | 0       | 0    |

--- Summary for K = 8 ---

Z.ar:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.9144 | 0.9733  | 0.9833 | 0.9799 | 0.9900  | 0.9972 |

beta.ar:

Modality 1 :

| Min.     | 1st Qu.  | Median   | Mean     | 3rd Qu.  | Max.     |
|----------|----------|----------|----------|----------|----------|
| 0.000000 | 0.006111 | 0.015556 | 0.029677 | 0.035556 | 0.755556 |

Modality 2 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 3 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 4 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 5 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006667 0.016667 0.030292 0.037778 0.552222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000e+00 0.000e+00 0.000e+00 3.820e-07 0.000e+00 2.167e-02
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.001133 0.000000 1.000000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0         0         0         0         0         0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0         0         0         0         0         0

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.9700 0.9911 0.9928 0.9926 0.9944 0.9994

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00500 0.01278 0.02548 0.02889 0.67778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.005556 0.014444 0.026026 0.031111 0.578889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 1.76e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0         0         0         0         0         0
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0         0         0         0         0         0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.

```

```
0.0000000 0.0000000 0.0000000 0.0000148 0.0000000 0.5444444
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5339 0.7894 0.8311 0.8104 0.8456 0.8817
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09167 0.13833 0.13373 0.17111 0.90111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09444 0.14167 0.13696 0.17556 0.69333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007778 0.025556 0.158427 0.184444 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0005556 0.0055556 0.0094444 0.0884393 0.0351389 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007778 0.031111 0.199776 0.260556 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5261 0.6944 0.7339 0.7229 0.7617 0.8111
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.00000 0.04611 0.09278 0.09781 0.13389 0.83611
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.04778 0.09667 0.10043 0.13889 0.63722
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0283333 0.3466888 0.8361111 1.0000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0027778 0.2025574 0.1670833 1.0000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.07167 0.82167 0.61118 0.96111 1.00000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.8839 0.9911 0.9939 0.9929 0.9961 0.9994
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.03056 0.06944 0.08765 0.11556 0.87611
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03222 0.07222 0.08907 0.12000 0.65222
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001767 0.0000000 0.5055556
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001906 0.0000000 0.0594444
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001198 0.0000000 0.0316667
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001001 0.0000000 0.1977778
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9600 0.9922 0.9944 0.9938 0.9961 1.0000
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01833 0.04583 0.06228 0.08278 0.82722
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02000 0.04778 0.06414 0.08667 0.75722
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.162e-05 0.000e+00 2.156e-01
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 1.353e-05 0.000e+00 1.722e-02
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 1.382e-05 0.000e+00 3.889e-03
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001802 0.0000000 1.0000000

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##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4994 0.6383 0.6900 0.6869 0.7400 0.8528
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.03333 0.04799 0.06556 0.81278
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01333 0.03556 0.04962 0.06889 0.53778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01101 0.00000 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.8189 0.9144 0.8828 0.9983 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.005507 0.000000 0.381111
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.0247 0.0000 0.8050
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5128 0.6889 0.7367 0.7326 0.7817 0.8711
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.009444 0.025000 0.039897 0.052222 0.857778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

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##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01000 0.02667 0.04089 0.05486 0.43222
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0004177 0.0000000 0.6261111
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.4128  0.6367  0.5873  0.7989  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0007404 0.0000000 0.6816667
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9189  0.9900  0.9928  0.9921  0.9944  0.9994
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008333 0.021111 0.041498 0.047778 0.866667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.022222 0.040849 0.051111 0.586667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0

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## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9144 0.9733 0.9833 0.9799 0.9900 0.9972
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006111 0.015556 0.029677 0.035556 0.755556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006667 0.016667 0.030292 0.037778 0.552222
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.820e-07 0.000e+00 2.167e-02
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.001133 0.000000 1.000000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9700 0.9911 0.9928 0.9926 0.9944 0.9994
##

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## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00500 0.01278 0.02548 0.02889 0.67778
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.014444 0.026026 0.031111 0.578889
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00e+00 0.00e+00 0.00e+00 1.76e-05 0.00e+00 1.00e+00
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0000148 0.0000000 0.5444444
##
## ‘ ‘ ‘

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## Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.